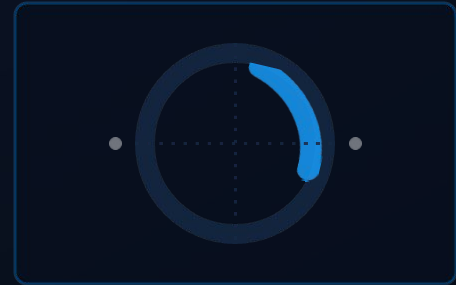


PROPOSAL

What Makes a Steam Game Popular?

Data Visualization Project Proposal

Prepared by: Sage Rudder



Project Overview



01 . Project Scope

Video games are one of the largest entertainment industries. This project explores success characteristics, investigating how **genre**, **price**, **user reviews**, **release year**, and **average playtime** relate to popularity.

02 . Dataset & Methods

The dataset contains thousands of Steam games with rich variables. This allows us to practice effective data visualization techniques, utilizing scatterplots, bar charts, box plots, heatmaps, and line charts.

03 . Ultimate Goal

The ultimate goal is to identify patterns that explain why some games become global phenomena, while demonstrating high-impact visual encoding and storytelling.

Dataset Preview

appid	name	release_date	english	developer	publisher	platforms	required_age	categories	genres	steamspy_tags	achievements	positive_ratings	negative_ratings	average_playtime	median_playtime	owners	price	
0	10	Counter-Strike	2000-11-01	1	Valve	Valve	windows;mac;linux	0	Multi-player;Online Multi-Player;Local Multi-P...	Action	Action;FPS;Multiplayer	0	124534	3339	17612	317	10000000-20000000	7.19
1	20	Team Fortress Classic	1999-04-01	1	Valve	Valve	windows;mac;linux	0	Multi-player;Online Multi-Player;Local Multi-P...	Action	Action;FPS;Multiplayer	0	3318	633	277	62	5000000-10000000	3.99
2	30	Day of Defeat	2003-05-01	1	Valve	Valve	windows;mac;linux	0	Multi-player;Valve Anti-Cheat enabled	Action	FPS;World War II;Multiplayer	0	3416	398	187	34	5000000-10000000	3.99
3	40	Deathmatch Classic	2001-06-01	1	Valve	Valve	windows;mac;linux	0	Multi-player;Online Multi-Player;Local Multi-P...	Action	Action;FPS;Multiplayer	0	1273	267	258	184	5000000-10000000	3.99
4	50	Half-Life: Opposing Force	1999-11-01	1	Gearbox Software	Valve	windows;mac;linux	0	Single-player;Multi-player;Valve Anti-Cheat en...	Action	FPS>Action;Sci-fi	0	5250	288	624	415	5000000-10000000	3.99
27070	1065230	Room of Pandora	2019-04-24	1	SHEN JIAWEI	SHEN JIAWEI	windows	0	Single-player;Steam Achievements	Adventure;Casual;Indie	Adventure;Indie;Casual	7	3	0	0	0	0-20000	2.09
27071	1065570	Cyber Gun	2019-04-23	1	Semyon Maximov	BekkerDev Studio	windows	0	Single-player	Action;Adventure;Indie	Action;Indie;Adventure	0	8	1	0	0	0-20000	1.69
27072	1065650	Super Star Blast	2019-04-24	1	EntwicklerX	EntwicklerX	windows	0	Single-player;Multi-player;Co-op;Shared/Split ...	Action;Casual;Indie	Action;Indie;Casual	24	0	1	0	0	0-20000	3.99
27073	1066700	New Yankee 7: Deer Hunters	2019-04-17	1	Yustas Game Studio	Alawar Entertainment	windows;mac	0	Single-player;Steam Cloud	Adventure;Casual;Indie	Indie;Casual;Adventure	0	2	0	0	0	0-20000	5.19
27074	1069460	Rune Lord	2019-04-24	1	Adept Studios GD	Alawar Entertainment	windows;mac	0	Single-player;Steam Cloud	Adventure;Casual;Indie	Indie;Casual;Adventure	0	4	0	0	0	0-20000	5.19

Screenshot of dataset columns and first/last few rows from Jupyter Notebook.

Dataset Variables

Column	Type	Description
appid	# Integer	Unique Steam application ID;
name	T Text	Name of the game;
release_date	📅 Date/Text	Date the game was released;
english	01 Binary	Indicates whether the game supports English;
developer	T Text	Game developer;
publisher	T Text	Company that published the game;
platforms	T Text	Supported operating systems (Windows, Mac, Linux);
required_age	T Integer	Minimum required player age;
categories	T Text	Steam feature categories (Single-player, Multiplayer, etc.);
genres	T Text	Primary game genres;
steamspy_tags	T Text	Community-generated gameplay tags;
achievements	# Integer	Number of Steam achievements;
positive_ratings	# Integer	Number of positive user reviews;
negative_ratings	# Integer	Number of negative user reviews;
average_playtime	# Integer	Average player playtime (minutes);
median_playtime	# Integer	Median player playtime (minutes);
owners	T Text	Estimated number of game owners;
price	. Decimal	Current game price in USD.



Dataset Information

 EDA-STEAM

**Dataset Name & Source**
EDA-Steam

**Author / Maintainer**
Ksawery Lejczak
Via Kaggle Notebook

**Kaggle URL**
kaggle.com/code/cloudy17/eda-steam/notebook

**Dimensions**
27,075 Rows & 18 Columns

 **Notable Dataset Variables**

Financial & Size

- Price, Owners

User Engagement

- Avg Playtime

Feedback Metrics

- Positive Ratings
- Negative Ratings

Categorical Tags

- Genres, Categories
- SteamSpy Tags

System Compatibility

- Platforms (Windows, Mac, Linux)

This mix of metrics allows for deep statistical correlation and multidimensional exploratory visualization.

Why This Dataset?



Large Volume

A massive dataset containing detailed information on thousands of games.



Variable Mix

Perfect blend of numerical metrics and diverse categorical features.



Highly Versatile

Highly suitable for applying a wide range of analytical visualization techniques.



Well-Documented

A trusted, clean, and thoroughly documented open-source Kaggle dataset with 0 null values.



Personal Interest

An engaging, gaming-centric topic that drives my personal motivation.



Rich Insights

Deep enough to successfully investigate multiple core research questions.

Initial Research Questions



**1. Does game price
relate to user
review ratings?**



**2. Which game
genres have the
highest average
playtime?**

These questions may evolve after exploratory data analysis.